

Mr. Qihua Dong (His/He/Him)

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Education

Northeastern University, Boston, USA

Sep 2023 - Now

- PhD in Computer Engineering under the supervision of Prof. Raymond Fu
- Interested topics: Multi-modal LLM, Computer Vision, Artificial Intelligence
- GPA: 3.96/4.00

City University of Hong Kong, Hong Kong SAR, China

Sep 2017 - Jul 2021

- BSc in Computer Science with Mathematics Minor; overall GPA 3.89/4.3 (Top 10% official proofed)
- Featured courses: Mathematical Analysis; Enhanced Calculus and Linear Algebra; Discrete Mathematics; Statistic Processes; Data Structures and Advanced Programming; Design and Analysis of Algorithms

Publications

- **Dong Qihua**, Kuo Yang, Handong Zhao, Yizhou Wang, Yun Fu. "Chain-of-Spatial-Thought: Emergent Spatial Reasoning in MLLM through Verifiable Rewards". *Ongoing*.
- **Dong Qihua**, Luis Figueroa, Handong Zhao, Kushal Kafle, Jason Kuen, Zhihong Ding, Scott Cohen, Yun Fu. "CoT Referring: Chain-of-Thought Grounding Improve Localization in Referring Expression Tasks." To be submitted to *Proc. ICCV, 2025*.
- **Dong Qihua**, Du Hao, Song Ying, Xu Yan, Liao Jing. "Preserving Tumor Volumes for Unsupervised Medical Image Registration". *Proc. ICCV, 2023*.
Paper link: <https://arxiv.org/pdf/2309.10153.pdf>
- He Ruozhen*, **Dong Qihua***, Lin Jiayin, Rynson Lau. "Weakly-Supervised Camouflaged Object Detection with Scribble Annotations". *Proc. AAAI, 2023*.
Paper link: <https://arxiv.org/pdf/2207.14083v2.pdf>
- Du Hao*, **Dong Qihua***, Xu Yan, Liao Jing. "Weakly-Supervised 3D Medical Image Segmentation using Geometric Prior and Contrastive Similarity". *IEEE Transactions on Medical Imaging, 2023*.
Paper link: <https://arxiv.org/pdf/2302.02125.pdf>
- Du Hao, **Dong Qihua**, Xu Yan, Liao Jing. "Contrastive Feature Extraction and Shared-Unique Clustered Attention for Multimodal 3D Medical Image Analysis". Submitted to *Journal of Biomedical and Health Informatics, 2024*.
- **Dong Qihua**, Yue Bai, Yizhou Wang, Yun Fu. "Seeing is Knowing: Advancing Semantic Understanding with MLLMs in Segmentation Tasks". *Ongoing*.

*: The authors with * contributed equally to the work

Research Experience

Multi-modal LLM in image understanding and grounding

May 2024 – Feb

- Supervisor: Luis Figueroa, Adobe
- Research brief: Developing an advanced multi-modal large language model (MLLM) for image segmentation tasks, focusing on segmentation on complex referring expression. Proposed a structured, chain-of-thought strategy enhancing model reasoning across modalities. The approach incorporates new annotations for existing datasets and a novel benchmark tailored for complex referring cases.

Multi-modal LLM in visual understanding and reasoning

Oct 2023 – Now

- Supervisor: Prof. Raymond Fu, Northeastern Univ.
- Research brief: Developed new multi-modal large language model (LLM) capable of visual grounding and reasoning, combined with spatial cognition. The final goal is building an intelligent agent capable of solving real-world problems with no costly finetuning but just some in-context examples. Visual understanding and thinking is critical, since it prevents hallucination and allow zero-shot generalization in complex tasks.

Preserving Tumor Volume in 3D Medical Image Registration

Sep 2022 – Mar 2023

- Supervisor: Prof Jing Liao, Prof Yan Xu, City Univ. of HK
- Research brief: Proposed a novel registration neural network that preserves the volume of tumors without the loss of alignment accuracy, allowing registration network to have wider clinical application. A new volume-preserving loss is designed specifically for the task.

Dynamic Transformer for 3D Medical Image Segmentation

Jun 2022 – Nov 2022

- Supervisor: Prof Jing Liao, Prof Yan Xu, City Univ. of HK
- Research brief: Designed an efficient and accurate transformer that adapts to medical image segmentation with dynamic token generation, in which only foreground-related tokens are further split and attended. Our method improves SOTA Dice scores of transformers on two popular datasets in medical images.

Weakly-Supervised 3D Medical Image Segmentation Framework

Feb 2022 - May 2022

- Supervisor: Prof Jing Liao, Prof Yan Xu, City Univ. of HK
- Research brief: Designed a new weakly supervised framework to segment medical images with bounding-box labels. The framework exploited the geometric prior of organs by using template organs in point-cloud form, which provides delicate shape guidance for segments. Besides, pretraining of the feature head was performed to learn contrasted features between objects and background. The work improves SOTA performance by 12.5% on Dice on two popular datasets in medical images.

Weakly-Supervised Camouflaged Object Detection

Dec 2021 - Jun 2022

- Supervisor: Prof Rynson Lau, City Univ. of HK
- Research brief: Proposed the first weakly-supervised task and dataset in camouflaged object detection. It contains the novel consistency loss that corrects an ignored bias in previous works. The feature loss leveraging semantic features to infer concealed objects. Our model outperforms SOTA methods on COD benchmarks with an average improvement of 11.0% on MAE.

Work Experience

Adobe Corp, CA, USA

Sep 2024 – Now

Mentor: Luis Figueroa (<https://luisf.me/>)

- Position: Extended Research Intern (Remote)
- Responsibilities: Research on referring image segmentation with Multimodal LLM.

Adobe Corp, CA, USA

May 2024 – Aug 2024

Mentor: Luis Figueroa (<https://luisf.me/>)

- Position: Research Intern (On-site)
- Responsibilities: Research on referring image segmentation with Multimodal LLM.

Texas A&M University, TX, USA

Oct 2022 – Jan 2023

- Supervisor: Prof Wenping Wang (<https://engineering.tamu.edu/cse/profiles/Wang-Wenping.html>)
- Position: Research Assistant (Remote)
- Responsibilities: Conducting research related to faces in computer vision and computer graphics.

City University of Hong Kong, Hong Kong SAR, China

Aug 2021 – Aug 2023

- Supervisor: Prof Jing Liao (<https://www.cityu.edu.hk/stfprofile/jingliao.htm>)
- Position: Research Assistant (Full-time)
- Responsibilities: Engaging the research projects in computer vision, the primary research domain is 3D Medical Image Analysis and Weakly-Supervised Learning.

City University of Hong Kong, Hong Kong SAR, China

Sep 2020- Feb 2021

- Supervisor: Prof Jianping Wang (<https://www.cs.cityu.edu.hk/~jianwang/>)
- Position: Student Research Assistant (Part-time)
- Responsibilities: Research works in computer science: the major research topics are related to system scheduling schemes for self-driving tasks.

Sik Sik Yuen, Hong Kong SAR, China

Sep 2019- Jun 2020

- Position: Full-stack Programmer (Intern)
- Responsibilities: Mobile phone application development project including AI functions and web service.

Honours and Scholarship

2021	First class honor , Bachelor's degree of City University of Hong Kong, HK
2019	Bronze Medal , The 2019 ICPC Asia Nanchang Regional Contest, Jiangxi Normal University, China
2017 - 2021	Dean's List , City University of Hong Kong, HK
2017 - 2021	Full Tuition Scholarship , City University of Hong Kong, HK

Skills

- Computer languages Python, C++, Matlab, R, Javascript, Java, CSS
- Tools and library PyTorch, Numpy, Detectron2, Skimage, mmDet, Monai, Scikit-learn,
- English skills TOEFL 110 (Nov 2022); GRE 330 (+3.5)